

Research papers

The nonlinear rainfall–quick flow relationships in a humid mountainous area: Roles of soil thickness and forest type

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ABSTRACT

The nonlinear behavior of hillslope quick flow (QF) in response to variations in soil thickness and forest types remains unclear due to uncertainties or unknown features in field studies, hindering accurate interpretation of their impact on QF. To this end, three artificial physical models with different mineral soil thickness (0 cm, shallow 50 cm, and thick 150 cm) beneath a 50 cm root zone for each forest (i.e., the broadleaf and coniferous forests) were constructed and intensively monitored. Results showed that the threshold-based QF was a function of the sum of rainfall amount and antecedent soil wetness, which was isolated by two breakpoints—the generation and rising thresholds—across all artificial physical models. These breakpoints indicated the storage thresholds for QF generation and the transition from slow to fast QF in humid forests. The broadleaf forest required more rainfall to reach the generation threshold due to higher litter interception capacity and soil water deficit by plant water uptake compared to the coniferous forest. Above the rising threshold phase, QF increased rapidly and linearly due to the establishment of flow path connectivity driven by soil water content, with the broadleaf forest showing a faster increase in QF than the coniferous forest. The dominant preferential/unsaturated flow in humid mountainous areas was also confirmed. Preferential flow paths were well-connected in the root zone and extended to the shallow mineral soil, resulting in an insignificant difference in mean QF between forest types. The thick mineral soil disrupted preferential flow path connectivity, leading to a greater amount of water being laterally diverted and retained within the soil. Consequently, a sharp decrease in QF was observed in hillslope with the root zone capping thick mineral soil for both forests. Our results highlighted the application of threshold-based theory and controlled experiments in enhancing the comprehension of the fundamental physical mechanism governing runoff processes in humid forests.

1. Introduction

Nonlinear or threshold behaviors are commonly observed due to alteration of the internal states or discontinuous boundary conditions in hydrological systems (Detty and McGuire, 2010a; McGuire and McDonnell, 2010; Zehe and Sivapalan, 2009). The nonlinear behaviors can occur across different climate zones, spatial scales, and geomorphic conditions (Ali et al., 2015; Saffarpour et al., 2016; Zehe et al., 2007). Previous studies identified different types and values of thresholds for

runoff processes, including rainfall depth (Lehmann et al., 2007; Tromp-van Meerveld and McDonnell, 2006), water table height (Detty and McGuire, 2010b), soil water content (Penna et al., 2011), and a sum of soil water content and rainfall depth (Saffarpour et al., 2016; Wang et al., 2022b). Below the threshold, little to no stormflow/quick flow (QF) was generated, whereas a significant increase in stormflow was recorded once the threshold was exceeded. The threshold response is typically attributed to the hydrological connectivity of the flow network, which is activated by subsurface matrix flow, preferential flow, and soil

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pipe flow (Uchida et al., 2005). The threshold theory has advanced the interpretation of factors controlling runoff generation and the analysis of underlying mechanisms regulating different stages of runoff, enhancing the process understanding of water movement across different spatial scales. While, studies have revealed considerable variability of QF amounts during the above threshold phase (Buttle et al., 2019; Fu et al., 2013; Zhang et al., 2021), hindering the accurate prediction and assessment of QF amounts during rainfall and limiting our insight into the hierarchical processes where the output from one process serves as input for another. Recently proposed multiple-phase threshold behaviors could well characterize the initial QF activation and the transition from slow to fast response of QF (Saffarpour et al., 2016; Wei et al., 2020), but the changes in threshold values due to continuous evolution in hillslope or catchment characteristics, such as soil thickness, vegetation cover, and soil properties, have received limited attention (Ali et al., 2015; Camporese et al., 2019; Camporese et al., 2014). This constrains a comprehensive understanding of the evolving hydrological behaviors associated with thresholds and limits deep insights into the dominant runoff mechanism.

Soil thickness and its spatial variation determine the geomorphology of hillslopes or catchments and associated water storage capacity, consequently impacting hydraulic gradient, water movement, and flow paths, thereby governing the runoff processes (Liu et al., 2013; Patton et al., 2018). Buttle et al. (2004) found a linear relationship between runoff and rainfall in thin soils, whereas thick soils exhibited a nonlinear behavior; Hopp and McDonnell (2009) argued that changes in soil thickness did not significantly affect subsurface runoff coefficients, despite their considerable impact on the degree and timing of hydrological connectivity. Zhang et al. (2022) found that surface and subsurface runoff contributions differed significantly between the shallow-soil plot and the deep-soil plot in the karst region. These findings consistently highlight the important and complex role of soil thickness in runoff processes, as variations in soil thickness not only change water storage but also divert flow paths within hydrological systems. This may potentially result in divergent runoff mechanisms even in areas with similar soil characteristics. However, many studies have treated soil as a uniform buffer without considering soil thickness-induced changes in preferential flow paths and associated characteristics (e.g., the continuity and connectivity). This is crucial given the dominant role that preferential flow plays in runoff processes both below and above the threshold phase in forests (Alaoui et al., 2011; Nanda and Safeeq, 2023).

In humid forests, tree roots and faunal activities are confined to relatively shallow soil layers, known as the root zone (Fan et al., 2017; Nespoulous et al., 2019). Below the root zone, there is a variable depth of mineral soil overlaying the bedrock, resulting from landscape evolution or geomorphic conditions (Hartmann et al., 2020; Laine-Kaulio et al., 2015). Numerous studies have found that soil water storage capacity is influenced by vegetation roots through the reconstruction of soil pores, and by soil thickness through the augmentation of pore space (Buttle et al., 2019). Additionally, root-branching networks, animal burrowing networks, and cracks promote preferential flow paths, facilitating the movement of water from the soil surface into subsurface (Lin, 2010; Nimmo, 2021). For instance, a study conducted on three neighboring hillslopes with intensive wells for measuring water table fluctuations found that the lower percentage of well activation at the grassland was due to a higher storage capacity caused by thicker soil, whereas the similar percentage of well activation between the coniferous and mixed forests was attributed to preferential flow paths (Bachmair et al., 2012). Moreover, soil thickness also influences the connectivity of preferential flow paths due to soil layering and loss of roots, which in turn affects the timing and quantity of QF, significantly impacting the relationship between rainfall and QF (Scaife et al., 2020). Meanwhile, limited knowledge exists regarding the extent to which an increase in soil thickness affects the connectivity of flow paths within forest types in combination with the threshold behaviors. Particularly, conducting such studies in the field is challenging due to the complex interactions among the

atmosphere, vegetation, and geomorphology on water retention and release within hydrological systems (Alaoui et al., 2011; Wang and Chen, 2021; Zhou et al., 2023), which make it difficult to provide an effective platform for investigating physical processes under controlled conditions.

Early studies have demonstrated that nuances of the surface and subsurface topography of hillslopes or catchments, even under similar climatic, pedologic, and geologic conditions, can lead to a wide variability in relevant processes and contrasting runoff mechanisms (Bachmair and Weiler, 2012; Jarecke et al., 2021). As a result, quantifying how studied field characteristics might explain differences in the threshold values becomes challenging, which in turn restricts our understanding, development, and application of rainfall-runoff models at various scales (Lehmann et al., 2007; Scaife and Band, 2017; Zehe and Sivapalan, 2009). Controlled experiments have the advantage of isolating irrelevant variables that might influence the results, thus clarifying the observed behavior (Volkman et al., 2017). Additionally, intercomparison is a key approach for developing a holistic picture of the factors driving hydrological system responses (Bachmair et al., 2012; Hazenberg et al., 2016).

In this study, we applied controlled experiments to explore the nonlinear behavior of QF under varying soil thickness in response to precipitation input in two contrasting humid forests. The main objectives of this work were to: (1) identify the variables related to the threshold behavior of the QF process in humid forests, (2) explore the roles of soil thickness and forest type on threshold behavior, and (3) gain insight into the dominant runoff mechanism(s) in humid mountainous forests.

2. Materials and methods

2.1. Study site and artificial physical models

This study focuses on two typical forest types in Mountain Gongga: coniferous and broadleaf forests. The forests are distributed along the elevation gradient of Mountain Gongga due to the dramatic rise in elevation over a short horizontal distance. Soil and vegetation surveys were conducted at 2280 m a.s.l. for the broadleaf forest and at 3020 m a.s.l. for the coniferous forest. The dominant species in the coniferous forest are *Abies fabri* and *Picea brachytyla*, whereas *Lithocarpus cleistocarpus* and *Cercidiphyllum japonicum* dominate in the broadleaf forest. The monsoon climate in this area is characterized by distinct wet (May–October) and dry (November–April) seasons, with 80 % of the total precipitation occurring during the wet seasons (Hu et al., 2018).

Previous work has indicated that 50 cm is a critical depth (Figure S1), below which plant roots almost disappear, except for anchoring taproots in both forest types. This is attributed to the humid climate, which consistently provides abundant moisture to the topsoil, eliminating the need for roots to grow deeper (Fan et al., 2017). The root zone has a higher soil organic content and saturated hydraulic conductivity (K_s) than the underlying mineral soil. The K_s decreases exponentially with depth, and we also find a higher saturated soil water content and residual soil water content of the root zone compared with the mineral soil (Wang et al., 2022a). Soil water retention curves (SWRCs) were determined by multistep-outflow experiments on undisturbed soil columns, where soil water content was measured at specific soil water potentials through Pressure Plate Extractor (Soil Moisture Equipment Corp. USA). The soil parameters related to the SWRCs were then modeled with a van Genuchten–Mualem model using the measured relationship between soil water content and soil water potential (Van Genuchten, 1980; Van Genuchten et al., 1991). Finally, we averaged the modeled soil parameters from replicates of the soil columns to obtain the effective soil parameters (Table 1). Additionally, more information on soil texture, root biomass, and rock fragment content can be found in our previous work (Wang et al., 2022a).

To allow hydrological systems under controlled conditions, we

Table 1

Soil parameters related to soil water retention curve obtained by multi-step outflow and modeled with van Genuchten–Mualem model (θ_s , θ_r , and θ_{fc} denote saturated water content, residual water content, and water content at field capacity; α is the inverse of the air-entry pressure; and n is a dimensionless parameter related to the curve shape).

Forest types	Depth (cm)	θ_s (cm ³ /cm ³)	θ_r (cm ³ /cm ³)	θ_{fc} (cm ³ /cm ³)	α (cm ⁻¹)	n (–)
Coniferous forest	10	0.66	0.18	0.31	0.07	1.42
	20	0.64	0.21	0.33	0.07	1.40
	50	0.49	0.14	0.25	0.05	1.42
	100	0.37	0.13	0.22	0.02	1.50
	200	0.35	0.12	0.20	0.02	1.52
Broadleaf forest	10	0.65	0.23	0.35	0.13	1.33
	20	0.63	0.24	0.36	0.09	1.34
	50	0.51	0.22	0.29	0.07	1.45
	100	0.33	0.07	0.16	0.08	1.32
	200	0.31	0.08	0.15	0.07	1.36

constructed three artificial physical models (PMs) for each forest. The PMs were composed of concrete boxes, each with an identical projected length of 3 m and a width of 1 m, set at a fixed inclination of 30° (Fig. 1a), and waterproof material was applied to the inside of the boxes to prevent leakage. Soils were moved to PMs in two ways: the root zone was cut into blocks and transported to PMs using 50 × 50 × 50 cm steel cubes with minimal disturbance; the poorly structured mineral soil (below 50 cm) was carefully collected and refilled into the concrete boxes at 10–20 cm intervals through manual dispersal and packing. A drainage layer (~3 cm) composed of gravel was placed on top of the concrete bed to facilitate water drainage (Fig. 1b). The PMs were situated at an open site near the Mountain Gongga Observation Station (~3000 m a.s.l.). The refilling process was completed by the end of September 2019, resulting in three distinct PMs representing the root zone (PM-0.5), the root zone capping thin mineral soil (PM-1), and thick mineral soil (PM-2), respectively.

2.2. Hydrometric observations

Rainfall input was recorded using three tipping buckets near the open site. Soil water potential was monitored by T4e probes (UMS GmbH, Germany) deployed at depths of 10, 20, and 50 cm in the PM-0.5; 10, 50, and 100 cm in the PM-1; and 10, 50, 100, and 200 cm in the PM-2. The probes at 200 cm encountered severe damage in the middle of the monitoring period, so subsequent data were substituted with monitoring data from 100 cm, where the soil water potential was synchronized and similar according to previous observation.

To drain potential outflows, a trench was positioned at the soil surface to collect surface runoff. Stainless-steel plates were placed at a depth of 20 cm in PM-0.5 and at a depth of 50 cm in both PM-1 and PM-2 to collect interflow. Subsurface flow along the concrete bed was drained through a hole in the lower part of the concrete box. All outflows were connected to tipping buckets via pipes for volume determination (Fig. 2b). Notably, no surface runoff or interflow was generated in any of the PMs due to the high K_s and abundant preferential flow paths. This finding agrees with field studies demonstrating that interflow was only detected at depths below 300 cm in both forests (Figure S2).

The tipping buckets and probes were connected to a data logger (CR1000, Campbell Scientific Inc, USA) via multiplexers to record the monitoring data at 15-min intervals. Before further analysis, the entire data set underwent screening. Early monitoring data collected before May 2020—eight months after the refilling process—was discarded to avoid the influence of soil disturbance. Soil water potential and outflow measurements taken at temperatures below 0 °C were excluded due to unreliable readings from frozen soils. Abnormally large oscillations between subsequent measurements during a rainfall event were visually checked and removed. Additionally, data with asynchronous rainfall-runoff relationships, equipment failures, or battery issues were also excluded.

2.3. Graphical hydrograph separation

We defined rainfall events based on a specific time interval without rainfall, which was set at 12 h. Delineated rainfall events with a total rainfall amount of less than 1 mm were excluded. To calculate QF, we subtracted the base flow, which was assumed to be equal to the initial discharge, from the total outflow (i.e., subsurface flow in this study). The end of the event was defined either when the discharge in the falling limb of the hydrograph equaled the initial discharge or by the onset of the next event if the discharge in the falling limb was not met. The initial discharge of an event hydrograph was defined as the discharge immediately before the first pulse of rainfall (Haga et al., 2005). Thus QF (mm) was calculated as follows:

$$QF = \sum_{t=T_i}^{t=T_e} (Q_t - Q_i) / (1000 \times A)$$

Here, QF represents the event quick flow (mm), Q_t (ml) denotes the subsurface flow during the t^{th} period, and Q_i denotes the initial discharge of the subsurface flow, T_i and T_e are the initial and end time of the event subsurface flow, and A (m²) is the projection area of the PM.

To characterize antecedent soil water storage, we used and calculated the depth equivalent antecedent soil wetness (ASW) by the

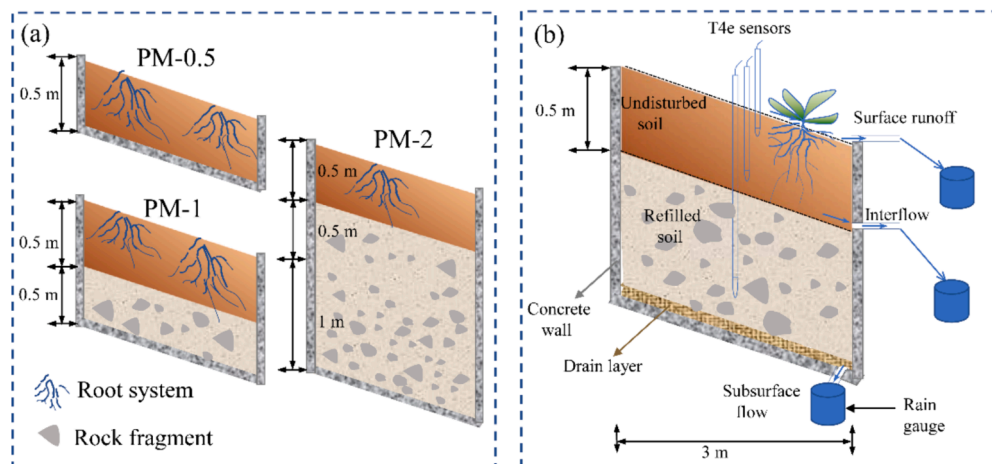


Fig. 1. Setups of (a) the artificial physical models and (b) monitoring network.

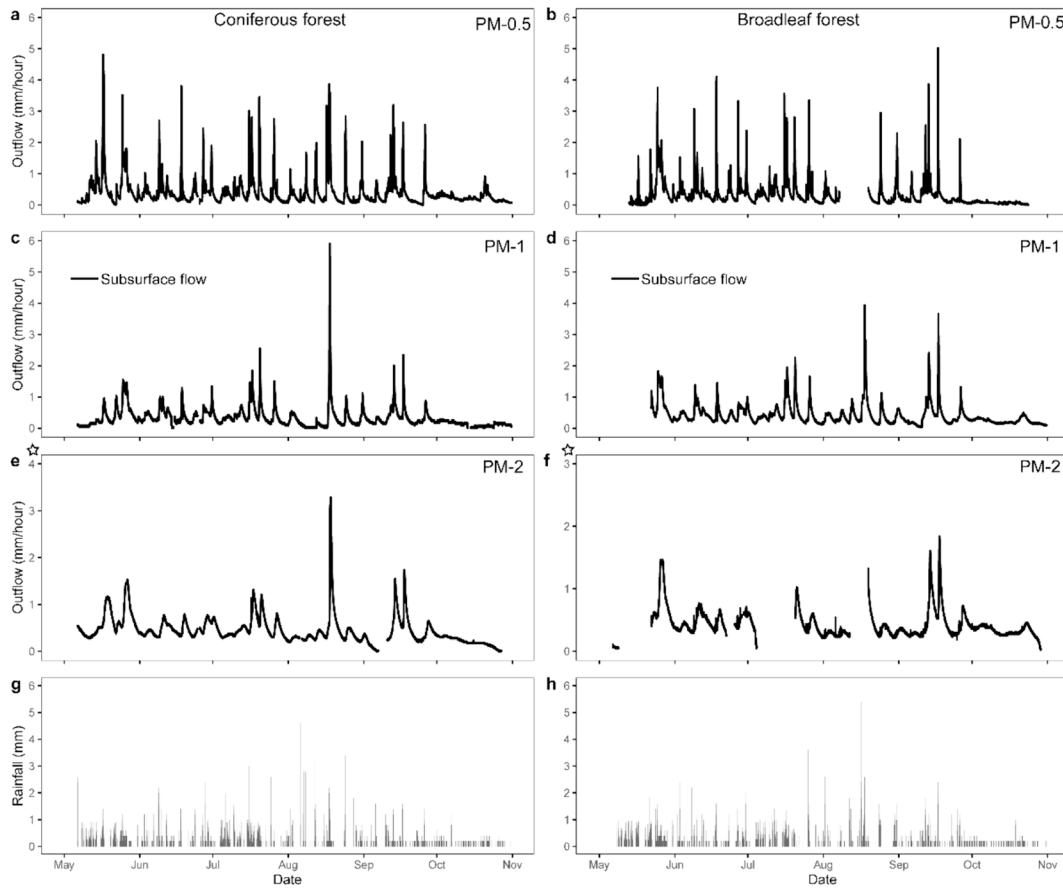


Fig. 2. Outflow and rainfall dynamics during the monitoring period for different PMs and forest types at 15-min intervals (Note that surface runoff and interflow did not occur during the monitoring period, and the star near y-axis denotes different scales).

following equation for different PMs:

$$\begin{aligned}
 ASW &= (\theta_{10} \times 15 + \theta_{20} \times 15) \times (1 - RF_1) + \theta_{50} \times 20 \times (1 - RF_2) \quad (\text{PM} - 0.5) \\
 ASW &= (\theta_{10} \times 30 + \theta_{50} \times 30) \times (1 - RF_1) + \theta_{100} \times 40 \times (1 - RF_2) \quad (\text{PM} - 1) \\
 ASW &= (\theta_{10} \times 30 + \theta_{50} \times 30) \times (1 - RF_1) + (\theta_{100} \times 60 + \theta_{200} \times 80) \times (1 - RF_2) \quad (\text{PM} - 2)
 \end{aligned}$$

Here, θ_i denotes the soil water content (SWC) measured at i th soil depth, and SWC was converted using SWRCs with measured soil water potential. RF_1 and RF_2 denote the mean rock fragment contents in different soil layers and were set at 7.5% and 12% in the coniferous forest, 6.5% and 15% in the broadleaf forest according to our previous work.

2.4. Approach for thresholds detection

To detect thresholds associated with QF, we aimed to define two types: generation and rising thresholds. The generation threshold represents the point at which a measurable output (i.e., subsurface flow in this study) occurs in 99 % of rainfall events. The rising threshold reflects a significant increase in QF above this threshold. To determine the thresholds, we used the piecewise regression analysis to calculate the threshold values (i.e., breakpoints) and derive linear slope parameters above and below the rising threshold. This approach benefits our understanding of the shift between slow and rapid QF. Piecewise regression

analysis was carried out using the “segmented” package in R software,

where breakpoints were automatically estimated based on the top-down approach and maximum likelihood approach (Muggeo, 2003).

3. Results

3.1. Rainfall–QF relationships

The total precipitation during the monitoring period (May 6th to November 6th) was 1684 mm, with a total of 60 rainfall events. Event precipitation (P) ranged from 1.60 to 128.00 mm, with an average of 28.46 mm and a mean rainfall intensity of 0.69 mm/hour, and 75 % of events had a total precipitation exceeding 4.5 mm. After data screening for PM-0.5, PM-1, and PM-2, the number of events with detected QF was 45, 30, and 27 in the coniferous forest, and 38, 30, and 18 in the broadleaf forest, respectively (Fig. 2).

The mean QF in the coniferous and broadleaf forests was 21.06 mm vs. 22.65 mm for PM-0.5, 20.62 mm vs. 26.28 mm for PM-1, and 17.74 mm vs. 20.62 mm for PM-2, respectively. QF was strongly correlated with P across all PMs ($R^2 > 0.77$, $p < 0.0001$), and linear fitting was applied to derive the slope parameters to present the QF yield capacity.

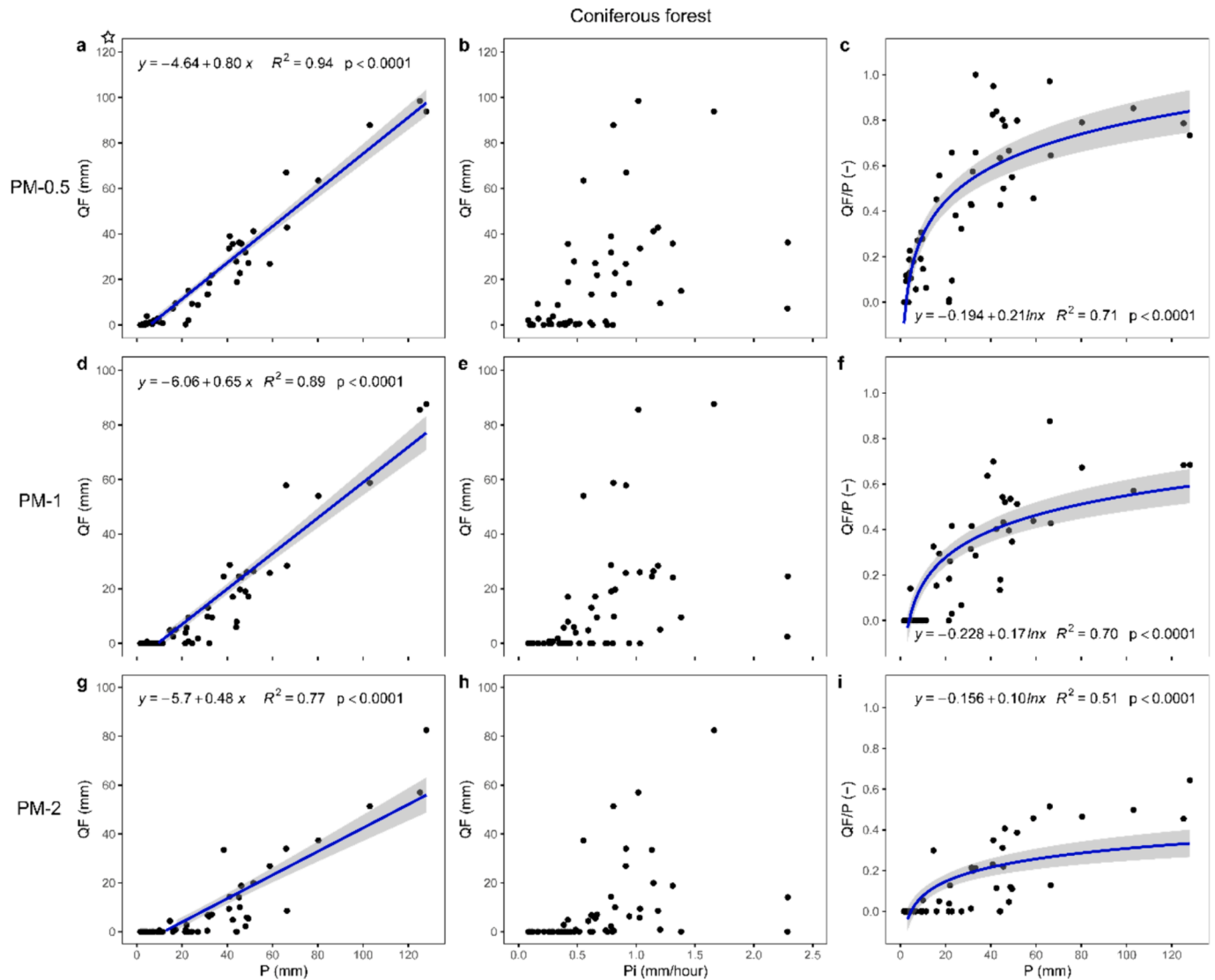


Fig. 3. Relationship between (a, d, g) event quick flow (QF) and precipitation (P), (b, e, h) QF and mean rainfall intensity (Pi), and (c, f, i) runoff coefficient (QF/P) and P in the coniferous forest (Blue solid line indicates the fitting curve, grey shaded area indicates 95% confidence interval). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

However, QF exhibited no evident correlation with rainfall intensity (P_i), particularly displaying a high variability pattern when P_i exceeded 1 mm/hour. The slope parameters for PM-0.5 and PM-1 in the broadleaf forest were higher than that in the coniferous forest, whereas the opposite trend was found for PM-2 (Figs. 3 and 4). The mean event quick flow coefficient (QF/P) in the coniferous and broadleaf forests was 0.46 vs. 0.49 for PM-0.5, 0.41 vs. 0.46 for PM-1, and 0.28 vs. 0.23 for PM-2, respectively. In each forest, there was no significant difference in QF/P between PM-0.5 and PM-1 ($p > 0.35$), while a significant difference was found when compared with PM-2 ($p < 0.001$). Additionally, there was an exponential relationship between QF/P and P across all PMs in both forests (Fig. 5c, f, i and Fig. 6c, f, i).

3.2. Threshold behaviors of QF

The threshold of rainfall depth for QF generation was not clearly defined across PMs, as both the presence and absence of QF were recorded within an extensive range of rainfall depth, especially in PMs with mineral soil (Figs. 3 and 4). The depth equivalent ASW revealed a distinct difference between forest types ($p < 0.01$). In the coniferous forest, the mean ASW values were 132.78 ± 6.52 mm (standard

deviation) for PM-0.5, 229.77 ± 10.22 mm for PM-1, and 395.68 ± 22.88 mm for PM-2. In the broadleaf forest, the mean ASW values were 102.28 ± 2.83 mm for PM-0.5, 189.64 ± 6.02 mm for PM-1, and 330.29 ± 21.83 mm for PM-2, respectively. Nevertheless, there was no evident relationship between QF and ASW, nor was a threshold response detected (Fig. 5a–c and 6a–c). Instead, the relationship between QF and ASW was highly variable across all PMs in both forests.

Meanwhile, when the ASW was combined with the precipitation, a distinct three-phase relationship was observed with QF across all PMs (Fig. 5d–f and 6d–f). The three-phase QF process was divided by the breakpoints of generation and rising thresholds (Figs. 5 and 6; Table 2). Between the generation and rising thresholds, the mean QF in the coniferous and broadleaf forests were 1.32 mm vs. 19.20 mm, 3.77 mm vs. 3.45 mm, and 6.55 mm vs. 7.92 mm for PM-0.5, PM-1, and PM-2, respectively; and the mean QF/P in the coniferous and broadleaf forests were 0.18 vs. 0.52, 0.26 vs. 0.20, and 0.17 vs. 0.18 for PM-0.5, PM-1, and PM-2, respectively. Above the rising threshold, the mean QF values in the coniferous and broadleaf forests were 28.05 mm vs. 71.6 mm for PM-0.5, 27.58 mm vs. 28.78 mm for PM-1, and 27.19 mm vs. 29.46 mm for PM-2, respectively; and QF/P was 0.53 vs. 0.83 for PM-0.5, 0.47 vs. 0.48 for PM-1, and 0.32 vs. 0.38 for PM-2 in the

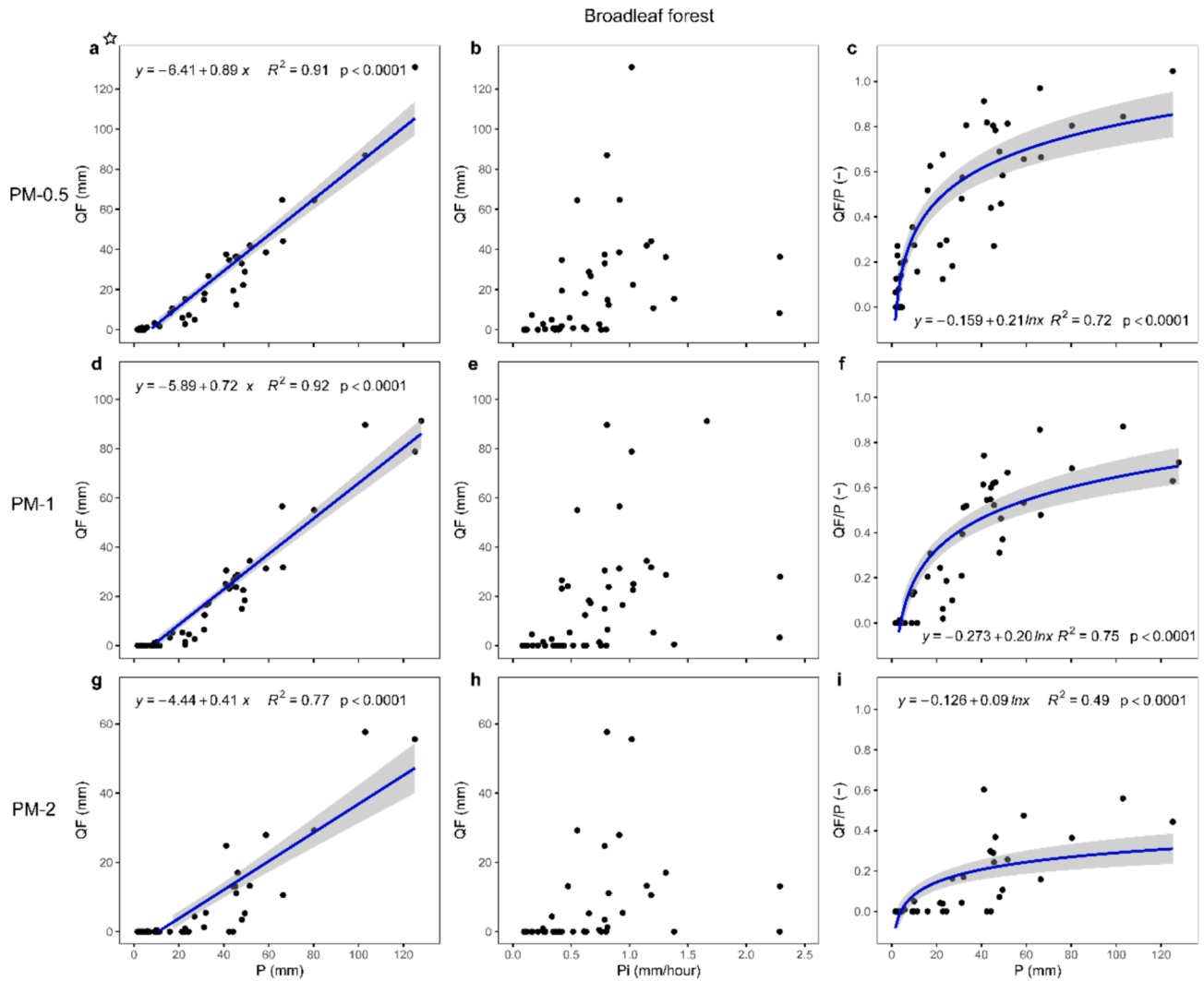


Fig. 4. Relationship between (a, d, g) event quick flow (QF) and precipitation (P), (b, e, h) QF and mean rainfall intensity (P_i), and (c, f, i) runoff coefficient (QF/P) and P in the broadleaf forest (Blue solid line indicates the fitting curve, grey shaded area indicates 95% confidence interval). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

coniferous forest and broadleaf forests, respectively.

Higher slope parameters were observed above the rising threshold phase in the broadleaf forest compared to the coniferous forest across all PMs. Specifically, the slope parameters were 0.83 vs. 1.27 for PM-0.5, 0.77 vs. 0.83 for PM-1, and 0.73 vs. 0.78 for PM-2 in the coniferous and broadleaf forests, respectively. However, the slope parameters between the generation and rising thresholds were extremely low for all PMs in both forest types, except for the PM-0.5 in the broadleaf forest, which had a slope parameter of 0.68. Notably, the slope parameters above the rising threshold decreased and underwent a narrower change as the thickness of mineral soil increased.

3.3. Soil water content and QF

SWC at rainfall onset was defined as the SWC at the last monitoring time before the rainfall event, and SWC at the generation and rising thresholds was estimated by the SWC at the time when the ASW+P reached the specific threshold values during the rainfall event. When comparing the SWC under different conditions, we applied a stringent significance level ($p < 0.01$) because all measurements were converted from soil water potential. The results showed that significant differences were observed among the SWC at different QF conditions ($p < 0.001$), except at a depth of 100 cm (Fig. 7). This may indicate a threshold value

of SWC in the root zone for QF generation and a shift to rapid QF. Besides, a comparison between forest types revealed that less water input was required to generate QF in the coniferous forest (Fig. 7).

4. Discussion

4.1. Role of forest types in threshold values and QF amount

Hydrological monitoring of QF revealed distinct differences in hydrological responses between the coniferous and broadleaf forests in terms of threshold values for QF generation and a shift to rapid QF (Table 2). To elucidate the differences between forest types, we compared the results of PM-0.5, which reflected the influence of hydrological processes from the vegetation and its root system. Firstly, a significantly lower threshold value of ASW+P for QF generation was identified in the broadleaf forest. Due to the presence of abundant preferential flow, subsurface flow can occur with a small amount of rainfall, resulting in a small proportion of rainfall compared to the depth equivalent ASW (P/ASW). This leads to the generation threshold being primarily determined by the ASW, which was higher in the coniferous forest than in the broadleaf forest (Fig. 7). Secondly, the discrepancy between ASW+P at the generation threshold and the mean ASW in the broadleaf forest (9.44 mm) was greater than that in the coniferous forest

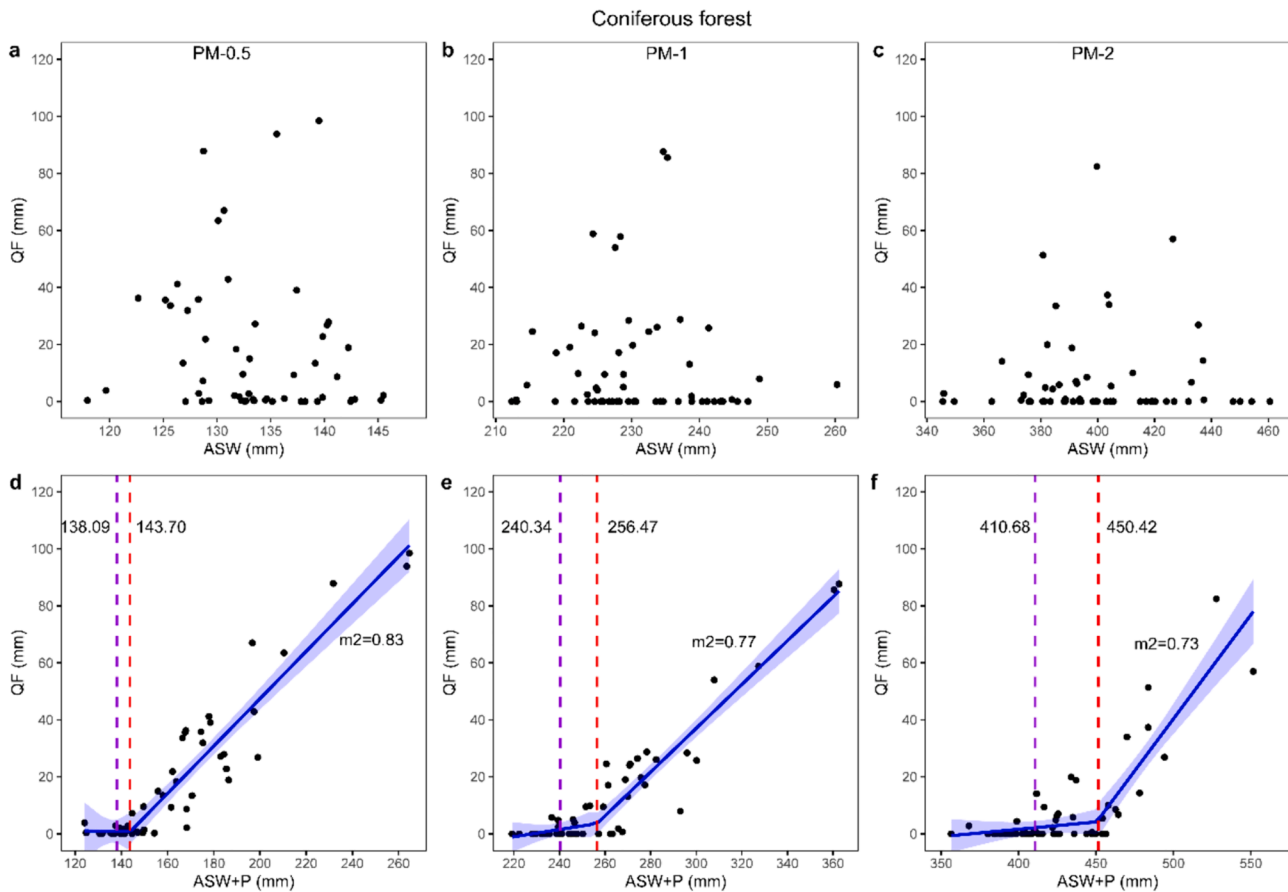


Fig. 5. Relationship between (a, b, c) event quick flow (QF) and ASW, and threshold relationship between (d, e, f) QF and ASW+P in the coniferous forest (Purple and red dashed lines indicate the generation and rising thresholds, respectively. Blue solid line indicates the piecewise regression for the variable of ASW+P, and grey shade area indicates the 95 % confidence interval). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(5.31 mm), indicating that a higher amount of rainfall was required to initiate subsurface flow in the broadleaf forest. This was attributed to the greater soil water deficit resulting from stronger water uptake by plants in the broadleaf forest. This aligned with the observed higher variability of soil water potential (Wang et al., 2023) and the measured higher evapotranspiration by the eddy covariance system (Hu et al., 2018) in the broadleaf forest compared to the coniferous forest. Additionally, the litter layer in the broadleaf forest exhibited a larger interception capacity compared to the coniferous forest due to the additional interception from bowl-shaped depressions of fallen leaves that facilitated the lateral spreading of rainfall (Du et al., 2019; Sato et al., 2004).

The coniferous forest exhibited a fast QF generation, whereas the broadleaf forest had a higher QF yield capacity, as indicated by the slope parameters (Figs. 3–6). Below the rising threshold, the slope parameter ($m_1 = 0.68$) in the broadleaf forest was much higher than that of the coniferous forest, indicating the establishment of vertical connectivity of flow paths in the broadleaf forest. However, the coniferous forest showed unpredictable QF below the rising threshold phase. Above the rising threshold, the slope parameter > 1 was observed in the broadleaf forest. This finding aligns with an obtained slope parameter of 2.36 in a deciduous forest, where the fast QF was attributed to the connectivity of more preferential flow paths and the contribution of more soil water (Buttle et al., 2019; Zhang et al., 2021). This is also consistent with a comparative study showing that the coniferous (spruce) forest had the least size of preferential flow paths and the lowest number of earthworms compared to the broadleaf (oak and alder) forest (Jačka et al., 2021). Therefore, a greater number of preferential flow paths in the broadleaf forest can drive and accumulate more rainwater in deep soils

within the same amount of time, leading to an increase in outflow amount.

4.2. Role of soil thickness on nonlinear behaviors

The nonlinear hydrological behaviors, resembling a hockey stick shape, were consistent across PMs, indicating that the underlying mechanism of the storage-discharge relationship—reflecting the retention and release of event water—remained unchanged (Kirchner, 2009; Zhang et al., 2022). Therefore, an increase in the mineral soil thickness would primarily affect the slope and intercept of the nonlinear relationship between QF and ASW+P (Ali et al., 2013; Kirchner, 2009). Notably, there was no significant difference in the QF/P between PM-0.5 and PM-1, which aligns with findings from a virtual experiment conducted on the Panola experimental hillslope. This experiment examined the impact of varying soil thickness, with a medium value of 120 cm, on flow metrics, and found that runoff coefficients were not greatly affected (Hopp and McDonnell, 2009). Two reasons may explain the phenomenon. On the one hand, a modest increase in mineral soil thickness caused a limited expansion of soil water storage deficit compared to the high event rainfall input; on the other hand, the long-range vertical connectivity (i.e., continuity) of preferential flow paths within the studied depth was relatively complete and contributed significantly to the QF amounts (Figure S3). This is line with the previous dye tracer experiments, which showed that rock fragments in the coarse-textured mineral soil resulted in continuous preferential flow paths to a depth of 110 cm and 95 cm in the coniferous forest and broadleaf forest, respectively (Wang et al., 2022a). As the thickness of mineral soil increased further,

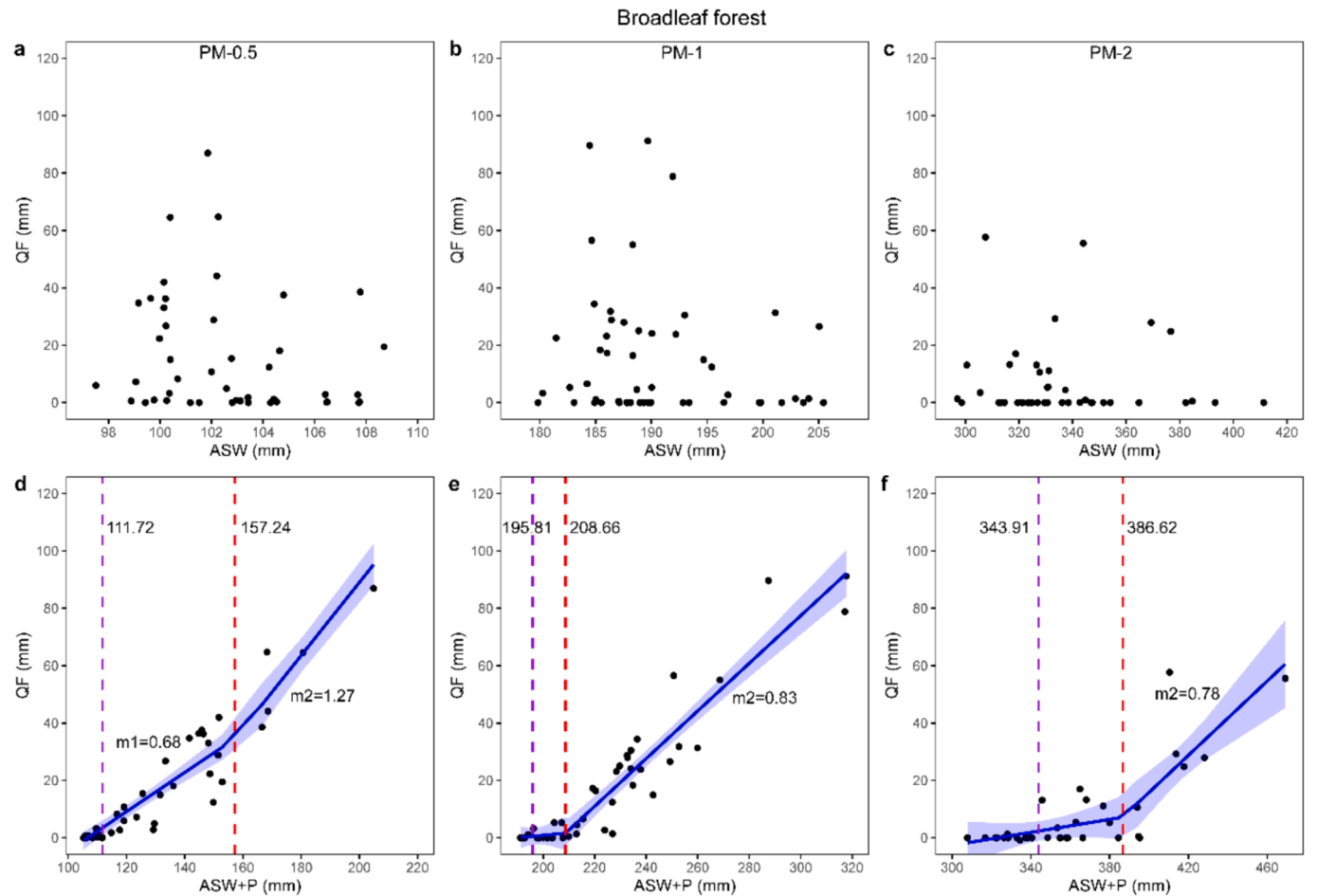


Fig. 6. Relationship between (a, b, c) event quick flow (QF) and ASW, and threshold relationship between (d, e, f) QF and ASW+P in the broadleaf forest (Purple and red dashed lines indicate the generation and rising thresholds, respectively. Blue solid line indicates the piecewise regression for the variable of ASW+P, and grey shade area indicates the 95 % confidence interval). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Summary from threshold analysis for different PMs and forest types.

Forest types	PM type	Generation threshold	Piecewise regression analysis				
			R ²	Rising threshold	SE	m ₁	m ₂
Coniferous forest	PM-0.5	138.09	0.88	143.70	4.81	0.01	0.83
	PM-1	240.34	0.89	256.47	4.90	0.13	0.77
	PM-2	410.68	0.77	450.42	5.52	0.05	0.73
Broadleaf forest	PM-0.5	111.72	0.93	157.24	8.66	0.68	1.27
	PM-1	195.81	0.90	208.66	4.90	0.08	0.83
	PM-2	343.91	0.70	386.62	11.23	0.05	0.78

Note: SE, threshold standard error of rising threshold;
m₁, slope parameter below the rising threshold;
m₂, slope parameter above the rising threshold;
R², coefficient of determination.

the long-range connectivity of preferential flow paths was lost due to the lack of biotic activities and the presence of soil layering. The water exchange between the preferential flow paths (e.g., macropores) and soil matrix became more pronounced, resulting in enhanced lateral water distribution and increased water retention within the soil (Nimmo, 2021). This result was supported by a study involving column experiments and numerical simulations, which showed that dead-end macropores reduced the flux into the main preferential flow paths and increased the lateral water transfer throughout the entire soil profile (Urbina et al., 2019). Consequently, there was a decrease in both QF/P and QF in the PM-2 compared to the other PMs. Meanwhile, the more

stable SWC content in the deeper mineral soil resulted in a limited expansion of the soil water storage deficit, contributing to the narrower decrease in the slopes above the rising threshold phase.

4.3. Runoff mechanisms in humid forests

The SWC consistently remained below saturation under various threshold conditions, with hardly observed soil saturation during the monitoring period (Figure S4). This indicated a dominant role of unsaturated/preferential flow in runoff processes in the studied area. When the rainfall fell on the ground, it was retained by the soil matrix and

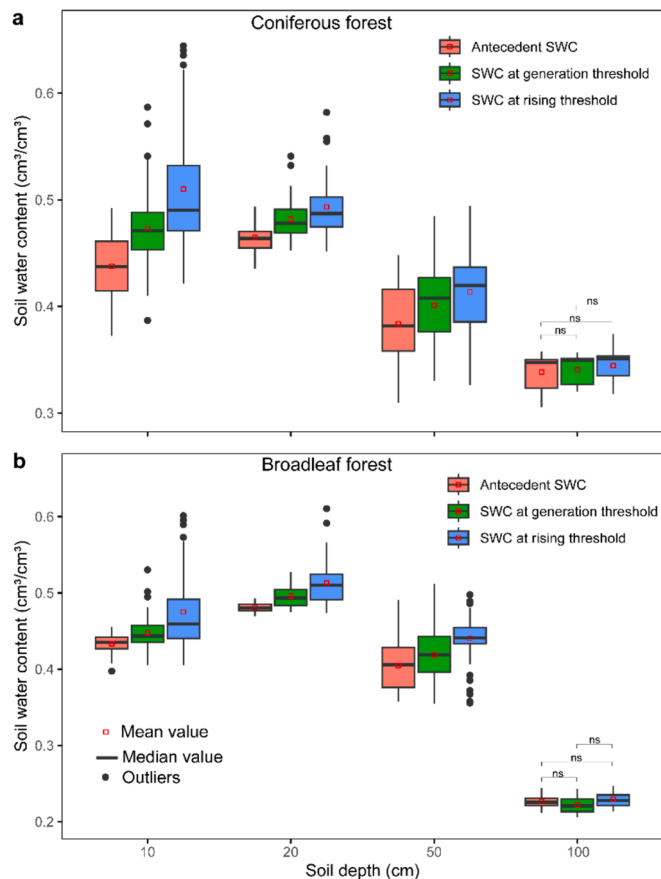


Fig. 7. Soil water content at rainfall onset (antecedent soil water content), generation, and rising threshold in the (a) coniferous and (b) broadleaf forest (The significant level was set at 0.01, “ns” denotes not significant).

replenished the soil water storage deficit during the first phase of the nonlinear behavior, making runoff unlikely to happen. Subsequent rainfall continuously recharged soil water, creating connectivity among segmented preferential flow paths within part of the hydrological system. However, this connectivity was unstable and disordered, leading to an unpredicted QF during the second phase. This phase was consistent with observations from the humid Lang Lang River Catchment in Australia, where runoff was even produced below the generation threshold value due to fast subsurface preferential flow paths (Saffarpour et al., 2016). As more rainfall entered the hydrological system, the degree of soil wetness progressively increased and a large preferential flow system was achieved through the self-organization processes of preferential flow paths. Consequently, rapid QF was generated and mainly controlled by rainfall (James and Roulet, 2007; Nieber and Sidle, 2010; Sidle et al., 2001; Zhang et al., 2021).

Therefore, the SWC at the rising threshold indicated a connected flow path for rapid QF. In this study, the SWC at 100 cm showed no significant difference ($p > 0.12$) among different hydrological phases. The mean SWC of 41 % in the coniferous forest and 44 % in the broadleaf forest at 50 cm was regarded as the threshold value of SWC for the connectivity of flow paths and the initiation of rapid QF (Fig. 7). The threshold value of SWC is similar to the 45 % found in Alpine grassland (Penna et al., 2011) and falls within the range of 41 %–46 % observed in the Tarrawarra catchment (Western and Grayson, 1998). However, it is higher than the 23 % recorded in a mature beech-maple forest (James and Roulet, 2007) and the 26 % found in a pine-oak forest (Farrick and Branfireun, 2014). Previous studies also found SWC threshold in near surface soil (James and Roulet, 2007; Penna et al., 2011) or the deeper soil (Farrick and Branfireun, 2014). These differences likely reflect the

variability in the soil properties, which depend on drainable porosity (calculated by subtracting the SWC at field capacity from the total porosity) to determine how much mobile soil water is available for drainage (Farrick and Branfireun, 2014; Scaife and Band, 2017), as well as by local climate, which influences the general status of SWC. Drainable porosity often decreases quickly with depth owing to the change in bulk density or soil texture, and a decrease in drainable porosity allows little water input to generate a measurable output (Weiler and McDonnell, 2004). As previously revealed by Farrick and Branfireun (2014) in a tropical dry forest, they found that when the SWC at the shallow soil layers (10, 30, and 50 cm) reached field capacity, excess rainwater can readily drain from shallow layers through the profile until the SWC at the deeper layer (100 cm) also attained field capacity. Once the unsaturated storage within the profile (0–100 cm) was met, the drainable water flowed laterally downslope and generated streamflow. In our study, a sharp increase in soil bulk density below 50 cm (Wang et al., 2022a), together with the temporarily stable high SWC (i.e., soil water potential) of the mineral soil (Wang et al., 2023) resulting from the humid climate, led to a high leaching potential (Table 1; Figure S4). Therefore, the SWC threshold of this study was identified at a shallow depth, where little rainfall is sufficient to drive the soil profile to a drainable/wet state. Under this condition, flow paths are organized into a larger flow network, facilitating rapid QF. Moreover, the slope parameters above the rising threshold decreased as the mineral soil thickness increased, and the difference in these slope parameters went narrower. This occurs because SWC tends to be more stable in deeper soil layers due to reduced influence from evapotranspiration, leading to a limited expansion of the soil water storage deficit.

4.4. Implications and future prospects

The absence of a unifying theory to connect or compare disparate process descriptions, coupled with the inherent uniqueness of study places, has spurred the development of increasing parameterized models aimed at representing diverse landscapes (Ali et al., 2013; Kirchner, 2006; Ross et al., 2021). However, the heterogeneity of landscape properties, as well as the limited transferability of findings from individual studies will persist and may even be exacerbated with enhanced observation capabilities and landscape changes over time. This has motivated a shift towards emergent properties, properties that cannot be predicted from the individual components but instead represent the interactions and feedbacks between the components in a complex system (McDonnell et al., 2007). Thresholds in runoff processes have been identified as an emergent property, reflecting that we can attain a rather good understanding of how a hydrological system works while being entirely ignorant of water particle physics. This will enhance the classification of runoff responses according to threshold theory, facilitating a better understanding of how runoff behaves under a wide variety of landscape and climate. Consequently, this will benefit the development of algorithms that embrace simplicity and applicability for predictive models, mitigating the issues of equifinality and our current reliance on overly calibrated models.

Threshold behaviors are dynamic in the context of long-term soil evolution, forest transformation, and climate change (Mekonnen et al., 2019; Zhang et al., 2024). The long-term nonequilibrium state between soil production and loss leads to changes in the spatial distribution of soil thickness, which is further exacerbated by climate change. Similarly, the forest transformation between deciduous and evergreen vegetation due to rising temperatures is also influenced by global climate change. Additionally, short-term strong disturbances, such as intensified floods, debris flow, and wildfire can impact the soil thickness and vegetation cover in a short time. These factors reflect the ubiquitous and dynamic changes of below-ground landscape elements. However, above-ground factors such as rainfall intensity and hydrological abstractions caused by evapotranspiration should also be taken into consideration.

Rainfall intensity thresholds, which are typically related to saturation excess flow and infiltration excess overland flow (Ali et al., 2015), received less attention compared to storage thresholds (Ross et al., 2021). Runoff mechanisms in natural or disturbed areas with limited infiltration capacity are heavily influenced by rainfall intensity. This influence has intensified due to the increased intra-event variability of rainfall intensity under global climate change (Labonté-Raymond et al., 2020), potentially leading to conversion of runoff mechanisms during a single rainfall event. In addition, the intensified seasonal variability of plant water uptake under global climate change can significantly alter soil water storage deficit and impact the threshold behaviors, leading to nonstationary threshold values. Hence, to adequately characterize hydrological responses of systems across various landscapes and climates, it is imperative to consider multiple highly variable and threshold-mediated processes in future studies.

5. Conclusion

Event-based analysis of the rainfall-quick flow relationship, varying by soil thickness and forest types, was conducted through controlled experiments to demonstrate how changes in water storage capacity and flow paths connectivity caused a threshold relationship between rainfall and hillslope quick flow. The results revealed that the nonlinear QF processes were related to the sum of antecedent soil wetness and rainfall amount in both coniferous and broadleaf forests. Forest types showed different litter interception capacity and plant water uptake, directly altering the soil water deficit and determining the rainfall amount for QF generation. While soil thickness expansion increased the soil water deficit and disturbed the flow paths connectivity, altering QF amount in response to rainfall events.

A three-phase threshold behavior with a hockey stick shape and two breakpoints—the generation and rising thresholds—was identified under varying soil thickness and forest types, indicating a consistent storage-discharge relationship in humid forests. For QF generation, the broadleaf forest required more rainfall due to higher litter interception and soil water deficit. Above the rising threshold, a linear relationship between QF and the sum of ASW and rainfall amount were observed, attributed to the establishment of preferential flow paths network, with the broadleaf forest showing higher increase in QF than the coniferous forest. Preferential flow paths are well-connected in the shallow root zone and extend to the shallow mineral soil due to the coarse-textured and stony soil in both forests. A thick mineral soil would disrupt flow path connectivity, enhancing the lateral diversion of rainwater and water retention within the soil, leading to a significant decrease in both the quantity and coefficient of QF in forests. This study highlighted the influence of soil thickness and forest types on threshold-based QF in humid mountainous areas and emphasized the need to shift the focus to factors controlling preferential flow and its connectivity, which have significant consequences for soil hydrology and threshold behaviors.

CRediT authorship contribution statement

Fei Wang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Genxu Wang:** Writing – review & editing, Supervision, Funding acquisition. **Junfang Cui:** Writing – review & editing, Resources, Funding acquisition. **Li Guo:** Writing – review & editing. **Xiangyu Tang:** Writing – review & editing, Methodology. **Ruxin Yang:** Investigation. **Jianqing Du:** Writing – review & editing. **Mohammad Sadegh Askari:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2024.131854>.

References

- Alaoui, A., Caduff, U., Gerke, H.H., Weingartner, R., 2011. Preferential Flow Effects on Infiltration and Runoff in Grassland and Forest Soils. *Vadose Zone J.* 10 (1), 367–377. <https://doi.org/10.2136/vzj2010.0076>.
- Ali, G., et al., 2013. Towards a unified threshold-based hydrological theory: necessary components and recurring challenges. *Hydrol. Process.* 27 (2), 313–318. <https://doi.org/10.1002/hyp.9560>.
- Ali, G., et al., 2015. Comparison of threshold hydrologic response across northern catchments. *Hydrol. Process.* 29 (16), 3575–3591. <https://doi.org/10.1002/hyp.10527>.
- Bachmair, S., Weiler, M., 2012. Hillslope characteristics as controls of subsurface flow variability. *Hydrol. Earth Syst. Sci.* 16 (10), 3699–3715. <https://doi.org/10.5194/hess-16-3699-2012>.
- Bachmair, S., Weiler, M., Troch, P.A., 2012. Intercomparing hillslope hydrological dynamics: Spatio-temporal variability and vegetation cover effects. *Hydrol. Earth Syst. Sci.* 48 (5), W05537. <https://doi.org/10.1029/2011WR011196>.
- Buttle, J.M., Dillon, P.J., Eerkes, G.R., 2004. Hydrologic coupling of slopes, riparian zones and streams: an example from the Canadian Shield. *J. Hydrol.* 287 (1), 161–177. <https://doi.org/10.1016/j.jhydrol.2003.09.022>.
- Buttle, J.M., Webster, K.L., Hazlett, P.W., Jeffries, D.S., 2019. Quickflow response to forest harvesting and recovery in a northern hardwood forest landscape. *Hydrol. Process.* 33 (1), 47–65. <https://doi.org/10.1002/hyp.13310>.
- Camporese, M., Penna, D., Borga, M., Paniconi, C., 2014. A field and modeling study of nonlinear storage-discharge dynamics for an Alpine headwater catchment. *Water Resour. Res.* 50 (2), 806–822. <https://doi.org/10.1002/2013wr013604>.
- Camporese, M., Paniconi, C., Putti, M., McDonnell, J.J., 2019. Fill and spill hillslope runoff representation with a Richards equation-based model. *Water Resour. Res.* 55 (11), 8445–8462. <https://doi.org/10.1029/2019WR025726>.
- Detty, J.M., McGuire, K.J., 2010a. Threshold changes in storm runoff generation at a till-mantled headwater catchment. *Water Resour. Res.* 46 (7), W07525. <https://doi.org/10.1029/2009wr008102>.
- Detty, J.M., McGuire, K.J., 2010b. Topographic controls on shallow groundwater dynamics: implications of hydrologic connectivity between hillslopes and riparian zones in a till mantled catchment. *Hydrol. Process.* 24 (16), 2222–2236. <https://doi.org/10.1002/hyp.7656>.
- Fan, Y., Miguez-Macho, G., Jobbágy, E.G., Jackson, R.B., Otero-Casal, C., 2017. Hydrologic regulation of plant rooting depth. *Proc. Natl. Acad. Sci.* 114 (40), 10572–10577. <https://doi.org/10.1073/pnas.1712381114>.
- Farrick, K.K., Branfireun, B.A., 2014. Soil water storage, rainfall and runoff relationships in a tropical dry forest catchment. *Water Resour. Res.* 50 (12), 9236–9250. <https://doi.org/10.1002/2014wr016045>.
- Fu, C., Chen, J., Jiang, H., Dong, L., 2013. Threshold behavior in a fissured granitic catchment in southern China: 1. Analysis of field monitoring results. *Water Resour. Res.* 49 (5), 2519–2535. <https://doi.org/10.1002/wrcr.20191>.
- Van Genuchten, M.v., Leij, F., Yates, S., 1991. The RETC code for quantifying the hydraulic functions of unsaturated soils. EPA Report 600/2-91/065, U.S. Salinity Laboratory, USDA-ARS, Riverside, California.
- Haga, H., et al., 2005. Flow paths, rainfall properties, and antecedent soil moisture controlling lags to peak discharge in a granitic unchanneled catchment. *Water Resour. Res.* 41 (12), W12410. <https://doi.org/10.1029/2005wr004236>.
- Hartmann, A., Semenova, E., Weiler, M., Blume, T., 2020. Field observations of soil hydrological flow path evolution over 10 millennia. *Hydrol. Earth Syst. Sci.* 24 (6), 3271–3288. <https://doi.org/10.5194/hess-24-3271-2020>.
- Hazenbergh, P., et al., 2016. Testing the hybrid-3-D hillslope hydrological model in a controlled environment. *Water Resour. Res.* 52, 1089–1107. <https://doi.org/10.1002/2015WR018106>.
- Hopp, L., McDonnell, J.J., 2009. Connectivity at the hillslope scale: Identifying interactions between storm size, bedrock permeability, slope angle and soil depth. *J. Hydrol.* 376 (3–4), 378–391. <https://doi.org/10.1016/j.jhydrol.2009.07.047>.

- Hu, Z., et al., 2018. Spatial-Temporal Patterns of Evapotranspiration Along an Elevation Gradient on Mount Gongga, Southwest China. *Water Resour. Res.* 54 (6), 4180–4192. <https://doi.org/10.1029/2018wr022645>.
- Jačka, L., Walmsley, A., Kovár, M., Frouz, J., 2021. Effects of different tree species on infiltration and preferential flow in soils developing at a clayey spoil heap. *Geoderma* 403, 115372. <https://doi.org/10.1016/j.geoderma.2021.115372>.
- James, A.L., Roulet, N.T., 2007. Investigating hydrologic connectivity and its association with threshold change in runoff response in a temperate forested watershed. *Hydrol. Process.* 21 (25), 3391–3408. <https://doi.org/10.1002/hyp.6554>.
- Jarecke, K.M., Bladon, K.D., Wondzell, S.M., 2021. The Influence of Local and Nonlocal Factors on Soil Water Content in a Steep Forested Catchment. *e2020WR028343* *Water Resour. Res.* 57 (5). <https://doi.org/10.1029/2020wr028343>.
- Kirchner, J.W., 2006. Getting the right answers for the right reasons: Linking measurements, analyses, and models to advance the science of hydrology. *Water Resour. Res.* 42 (3), W03S04. <https://doi.org/10.1029/2005wr004362>.
- Kirchner, J.W., 2009. Catchments as simple dynamical systems: Catchment characterization, rainfall-runoff modeling, and doing hydrology backward. *Water Resour. Res.* 45 (2), W02429. <https://doi.org/10.1029/2008WR006912>.
- Labonté-Raymond, P.-L., Pabst, T., Bussièr, B., Bresson, E., 2020. Impact of climate change on extreme rainfall events and surface water management at mine waste storage facilities. *J. Hydrol.* 590, 125383 <https://doi.org/10.1016/j.jhydrol.2020.125383>.
- Laine-Kaulio, H., Backnäs, S., Koivusalo, H., Laurén, A., 2015. Dye tracer visualization of flow patterns and pathways in glacial sandy till at a boreal forest hillslope. *Geoderma* 259–260, 23–34. <https://doi.org/10.1016/j.geoderma.2015.05.004>.
- Lehmann, P., Hinz, C., McGrath, G., Tromp-van Meerveld, H., McDonnell, J.J., 2007. Rainfall threshold for hillslope outflow: an emergent property of flow pathway connectivity. *Hydrology and Earth System Sciences* 11 (2), 1047–1063. <https://doi.org/10.5194/hess-11-1047-2007>.
- Lin, H., 2010. Linking principles of soil formation and flow regimes. *J. Hydrol.* 393 (1–2), 3–19. <https://doi.org/10.1016/j.jhydrol.2010.02.013>.
- Liu, J., Chen, X., Lin, H., Liu, H., Song, H., 2013. A simple geomorphic-based analytical model for predicting the spatial distribution of soil thickness in headwater hillslopes and catchments. *Water Resour. Res.* 49 (11), 7733–7746. <https://doi.org/10.1002/2013wr013834>.
- McDonnell, J.J., et al., 2007. Moving beyond heterogeneity and process complexity: A new vision for watershed hydrology. *Water Resour. Res.* 43 (7), W07301. <https://doi.org/10.1029/2006WR005467>.
- McGuire, K.J., McDonnell, J.J., 2010. Hydrological connectivity of hillslopes and streams: Characteristic time scales and nonlinearities. *Water Resour. Res.* 46 (10), W10543. <https://doi.org/10.1029/2010wr009341>.
- Mekonnen, Z.A., Riley, W.J., Randerson, J.T., Grant, R.F., Rogers, B.M., 2019. Expansion of high-latitude deciduous forests driven by interactions between climate warming and fire. *Nat. Plants* 5 (9), 952–958. <https://doi.org/10.1038/s41477-019-0495-8>.
- Muggeo, V.M.R., 2003. Estimating regression models with unknown break-points. *Statistic in Medicine* 22 (19), 3055–3071. <https://doi.org/10.1002/sim.1545>.
- Nanda, A., Safeeq, M., 2023. Threshold controlling runoff generation mechanisms in Mediterranean headwater catchments. *J. Hydrol.* 620, 129532 <https://doi.org/10.1016/j.jhydrol.2023.129532>.
- Nespoulous, J., et al., 2019. Tropical forest structure and understorey determine subsurface flow through biopores formed by plant roots. *Catena* 181, 104061. <https://doi.org/10.1016/j.catena.2019.05.007>.
- Nieber, J.L., Sidle, R.C., 2010. How do disconnected macropores in sloping soils facilitate preferential flow? *Hydrological Processes* 24 (12), 1582–1594. <https://doi.org/10.1002/hyp.7633>.
- Nimmo, J.R., 2021. The processes of preferential flow in the unsaturated zone. *Soil Sci. Soc. Am. J.* 85 (1), 1–27. <https://doi.org/10.1002/saj2.20143>.
- Patton, N.R., Lohse, K.A., Godsey, S.E., Crosby, B.T., Seyfried, M.S., 2018. Predicting soil thickness on soil mantled hillslopes. *Nature Communications* 9 (1), 3329. <https://doi.org/10.1038/s41467-018-05743-y>.
- Penna, D., Tromp-van Meerveld, H.J., Gobbi, A., Borga, M., Dalla Fontana, G., 2011. The influence of soil moisture on threshold runoff generation processes in an alpine headwater catchment. *Hydrol. Earth Syst. Sci.* 15 (3), 689–702. <https://doi.org/10.5194/hess-15-689-2011>.
- Ross, C.A., Ali, G.A., Spence, C., Courchesne, F., 2021. Evaluating the Ubiquity of Thresholds in Rainfall-Runoff Response Across Contrasting Environments. *e2020WR027498* *Water Resour. Res.* 57 (1). <https://doi.org/10.1029/2020wr027498>.
- Saffarpour, S., Western, A.W., Adams, R., McDonnell, J.J., 2016. Multiple runoff processes and multiple thresholds control agricultural runoff generation. *Hydrol. Earth Syst. Sci.* 20 (11), 4525–4545. <https://doi.org/10.5194/hess-20-4525-2016>.
- Scaife, C.I., Band, L.E., 2017. Nonstationarity in threshold response of stormflow in southern Appalachian headwater catchments. *Water Resour. Res.* 53 (8), 6579–6596. <https://doi.org/10.1002/2017wr020376>.
- Scaife, C.I., Singh, N.K., Emanuel, R.E., Miniat, C.F., Band, L.E., 2020. Non-linear quickflow response as indicators of runoff generation mechanisms. *Hydrol. Process.* 34 (13), 2949–2964. <https://doi.org/10.1002/hyp.13780>.
- Sidle, R.C., Noguchi, S., Tsuboyama, Y., Laursen, K., 2001. A conceptual model of preferential flow systems in forested hillslopes: evidence of self-organization. *Hydrol. Process.* 15, 1675–1692. <https://doi.org/10.1002/hyp.233>.
- Tromp-van Meerveld, H.J., McDonnell, J.J., 2006. Threshold relations in subsurface stormflow: 1. A 147-storm analysis of the Panola hillslope. *Water Resour. Res.* 42 (2), W02410. <https://doi.org/10.1029/2004wr003778>.
- Uchida, T., Tromp-van Meerveld, I., McDonnell, J.J., 2005. The role of lateral pipe flow in hillslope runoff response: an intercomparison of non-linear hillslope response. *J. Hydrol.* 311 (1), 117–133. <https://doi.org/10.1016/j.jhydrol.2005.01.012>.
- Urbina, C.A.F., et al., 2019. Water Flow in Soils with Heterogeneous Macropore Geometries. *Vadose Zone J.* 18 (1), 1–17. <https://doi.org/10.2136/vzj2019.02.0015>.
- Van Genuchten, M.T., 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am. J.* 44 (5), 892–898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>.
- Volkman, T.H.M. et al., 2017. Controlled Experiments of Hillslope Coevolution at the Biosphere 2 Landscape Evolution Observatory: Toward Prediction of Coupled Hydrological, Biogeochemical, and Ecological Change. In: Jiu-Fu Liu, W.-Z.G. (Ed.), *Hydrology of Artificial and Controlled Experiments*. InTech. DOI:10.5772/intechopen.72325.
- Wang, F., et al., 2022a. Preferential flow patterns in forested hillslopes of east Tibetan Plateau revealed by dye tracing and soil moisture network. *Eur. J. Soil Sci.* 73 (4), 13294. <https://doi.org/10.1111/ejss.13294>.
- Wang, F., et al., 2023. Hillslope-scale variability of soil water potential over humid alpine forests: Unexpected high contribution of time-invariant component. *J. Hydrol.* 617, 129036 <https://doi.org/10.1016/j.jhydrol.2022.129036>.
- Wang, J., Chen, L., 2021. The effect of hillslope geometry on Hortonian rainfall-infiltration-runoff processes. *J. Hydrol.* 594, 125962 <https://doi.org/10.1016/j.jhydrol.2021.125962>.
- Wang, S., Yan, Y., Fu, Z., Chen, H., 2022b. Rainfall-runoff characteristics and their threshold behaviors on a karst hillslope in a peak-cluster depression region. *J. Hydrol.* 605, 127370 <https://doi.org/10.1016/j.jhydrol.2021.127370>.
- Wei, L., Qiu, Z., Zhou, G., Kinouchi, T., Liu, Y., 2020. Stormflow threshold behaviour in a subtropical mountainous headwater catchment during forest recovery period. *Hydrol. Process.* 34 (8), 1728–1740. <https://doi.org/10.1002/hyp.13658>.
- Weiler, M., McDonnell, J., 2004. Virtual experiments: a new approach for improving process conceptualization in hillslope hydrology. *J. Hydrol.* 285 (1–4), 3–18. [https://doi.org/10.1016/s0022-1694\(03\)00271-3](https://doi.org/10.1016/s0022-1694(03)00271-3).
- Western, A.W., Grayson, R.B., 1998. The Tarawarra Data Set: Soil moisture patterns, soil characteristics, and hydrological flux measurements. *Water Resour. Res.* 34 (10), 2765–2768. <https://doi.org/10.1029/98wr01833>.
- Zehe, E., Elsenbeer, H., Lindenmaier, F., Schulz, K., Blöschl, G., 2007. Patterns of predictability in hydrological threshold systems. *Water Resour. Res.* 43 (7), W07434. <https://doi.org/10.1029/2006wr005589>.
- Zehe, E., Sivapalan, M., 2009. Threshold behaviour in hydrological systems as (human) geo-ecosystems: manifestations, controls, implications. *Hydrol. Earth Syst. Sci.* 13 (7), 1273–1297. <https://doi.org/10.5194/hess-13-1273-2009>.
- Zhang, G., et al., 2021. Stormflow generation in a humid forest watershed controlled by antecedent wetness and rainfall amounts. *J. Hydrol.* 603, 127107 <https://doi.org/10.1016/j.jhydrol.2021.127107>.
- Zhang, Y., Hartemink, A.E., Vanwalleghem, T., Bonfatti, B.R., Moen, S., 2024. Climate and land use changes explain variation in the A horizon and soil thickness in the United States. *Communications Earth & Environment* 5 (1), 129. <https://doi.org/10.1038/s43247-024-01299-6>.
- Zhang, J., Wang, S., Fu, Z., Chen, H., Wang, K., 2022. Soil thickness controls the rainfall-runoff relationship at the karst hillslope critical zone in southwest China. *J. Hydrol.* 609, 127779 <https://doi.org/10.1016/j.jhydrol.2022.127779>.
- Zhou, Z., et al., 2023. The deterioration and restoration of dried soil layers: New evidence from a precipitation manipulation experiment in an artificial forest. *J. Hydrol.* 625, 130087 <https://doi.org/10.1016/j.jhydrol.2023.130087>.